

AESAR — MASTER IMPLEMENTATION CONTRACT

Autonomous Ecosystem Scouting & Analysis Rover for Non-Pesticidal Management

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Autonomous Ecosystem Scouting & Analysis Rover for Non-Pesticidal Management

Team Engineering Contract v2.1 — Hackathon Source of Truth

Repository: AESAR

Base platform: Existing working ARACHNID autonomous rover

Status: TEAM SOURCE OF TRUTH

No-assumption rule: If a value, pin, protocol, threshold, coefficient, sensor, API field, state, or behavior is not explicitly defined in this contract or an authoritative referenced project document, it must not be invented. Mark it TBD, document the gap, and notify the integration owner.

1. AUTHORITY AND SOURCE OF TRUTH

This contract consolidates the current AESAR implementation baseline from the project's AESAR technical specification, actual hardware/communication audit, direct-Wi-Fi architecture, and ARACHNID integration documents.

1.1 Authority order

When sources conflict, use this order:

1. Actual tested hardware/firmware and latest direct-Wi-Fi hardware audit.
2. This contract.
3. AESAR research paper / Master Implementation Contract v2.0.0.
4. ARACHNID direct-Wi-Fi architecture.
5. Older ARACHNID ROS2 documents.
6. General engineering assumptions are not authoritative.

Older ROS2 documents are historical/reference material where they conflict with the direct-Wi-Fi implementation.

1.2 Prohibited assumptions

Do not assume:

- wheel encoders exist;
- ROS2 is in the physical path;
- the current Drive ESP32 accepts JSON velocity commands;
- the ToF is fixed rather than servo-scanned;
- frontend can communicate directly with ESP32;
- a sensor exists merely because an old document mentions it;
- a GPIO is available without hardware verification;
- an undocumented PDR/AMRI weight or threshold;
- simulated navigation equals physically validated navigation;
- MPU6050 provides accurate global X/Y localization;

- a model is trained or accurate unless actually tested.

2. PROJECT DEFINITION

AESAR automates a 10-plant diagonal CESA/AESA-style ecological scouting workflow for Non-Pesticidal Management.

The system combines:

- ground-level under-canopy sensing;
- multi-angle plant imaging;
- pest/defender detection;
- soil and microclimate measurements;
- Pest-to-Defender Ratio (PDR);
- Abiotic Microclimate Risk Index (AMRI);
- AESA/NPM decision logic;
- station field cards;
- mapping and telemetry;
- mission orchestration.

2.1 ARACHNID baseline

ARACHNID supplies:

- four-wheel differential-drive rover;
- dual ESP32 controllers;
- motors and dual BTS7960 drivers;
- 5 ultrasonic channels;
- VL53L0X ToF;
- servo-mounted ToF scan;
- existing mapping/navigation;
- Wi-Fi/TCP;
- tested low-level movement.

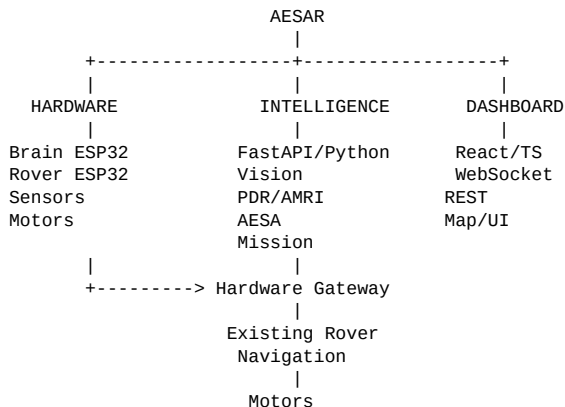
2.2 AESAR additions

AESAR adds:

- MPU6050 heading assistance;
- agricultural/environmental sensing;
- camera interface;
- YOLO vision;
- pest/defender classification;
- PDR;
- AMRI;
- AESA decision engine;
- NPM recommendation;
- field cards;
- 10-station mission;

- ecological/semantic presentation.

3. SYSTEM ARCHITECTURE



3.1 Layer 1 — Embedded

Execution:

- Brain ESP32 #2;
- Rover ESP32 #1;
- physical sensors/actuators.

3.2 Layer 2 — Autonomy and AESA intelligence

Execution:

- laptop/edge computer;
- Python 3.10+;
- FastAPI.

3.3 Layer 3 — Dashboard

Execution:

- browser;
- React 18 + TypeScript + Vite.

Browser connects only to backend.

4. HARDWARE CONTRACT

4.1 Brain ESP32

Known tested/current functions:

GPIO	Function
19	UART TX → Rover RX
18	UART RX ← Rover TX
14	ToF servo

GPIO	Function
21	VL53L0X SDA
22	VL53L0X SCL
25	Ultrasonic shared TRIG
32	Front-center ultrasonic ECHO
27	45°-left ultrasonic ECHO
35	45°-right ultrasonic ECHO
34	Left ultrasonic ECHO
33	Right ultrasonic ECHO

MPU6050:

- SDA → GPIO21;
- SCL → GPIO22;
- 3.3V;
- GND;
- shares I2C with VL53L0X.

Any additional agricultural sensor pin allocation must be verified against the physical assembly before use.

4.2 Rover ESP32

GPIO	Function
4	Left BTS7960 RPWM
17	Left BTS7960 LPWM
23	Left R_EN
5	Left L_EN
22	Right BTS7960 RPWM
21	Right BTS7960 LPWM
18	Right R_EN
19	Right L_EN
26	UART RX ← Brain TX
25	UART TX → Brain RX

Preserve the tested motor polarity/direction mapping.

4.3 UART

Brain ↔ Rover:

- 115200 baud;
- 8 data bits;
- no parity;
- 1 stop bit.

4.4 Motor protocol

Current physical command protocol:

```
FORWARD\n
BACKWARD\n
LEFT\n
RIGHT\n
STOP\n
```

The current integration must not send JSON motor packets.

Do not send JSON and ASCII motor commands back-to-back.

A future velocity protocol is outside the initial critical path.

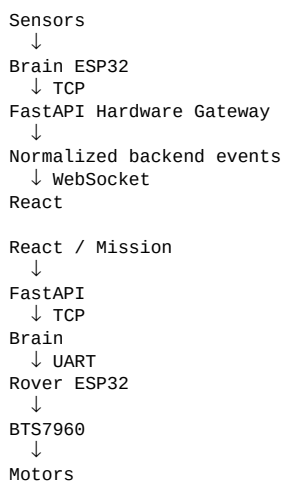
5. NETWORK CONTRACT

Current tested Brain network:

- SSID: ARACHNID;
- password: arachnid123;
- IP: 192.168.4.1;
- TCP port: 80.

The AESAR specification contains AESAR_ROVER in some sections, but the tested firmware/audit baseline is authoritative for the physical demo.

5.1 Data direction



6. TELEMETRY CONTRACT

6.1 Canonical logical telemetry

```
{
  "type": "sensor_telemetry",
  "timestamp_ms": 123456789,
  "seq": 42,
  "ultrasonic": {
    "front_mm": 420,
    "front_left_mm": 510,
    "front_right_mm": 490,
    "left_mm": 1200,
    "right_mm": 1150
  },
  "tof_scan": {
    "angle_deg": 118,
    "distance_mm": 380
  },
  "imu": {
    "accel_x_mps2": 0.0,
    "accel_y_mps2": 0.0,
    "accel_z_mps2": 9.81,
    "gyro_x_rads": 0.0,
    "gyro_y_rads": 0.0,
    "gyro_z_rads": 0.0,
    "yaw_deg": 0.0
  },
  "abiotic_sensors": {
    "soil_moisture_percent": null,
    "canopy_temp_c": null,
    "relative_humidity_percent": null
  }
}
```

```

    },
    "hardware_status": {
        "brain_wifi_rssi": null,
        "rover_uart_connected": true,
        "safety_stop_latched": false
    }
}

```

Fields not physically implemented must not be fabricated.

6.2 Units

Hardware:

- range = millimetres;
- acceleration = m/s²;
- angular velocity = rad/s.

Software:

- position = metres;
- navigation yaw = radians;
- linear velocity = m/s;
- angular velocity = rad/s.

UI may display degrees for human readability.

6.3 Invalid values

The old firmware may use 999 as an invalid/disconnected range sentinel.

Normalized software representation:

```
-1 = invalid/unavailable
```

Never interpret invalid range data as clear space.

7. TOF CONTRACT

Hardware:

- VL53L0X;
- servo-mounted;
- expected scan range approximately 30°–150°;
- expected step approximately 2°.

Logical sample:

```

{
  "angle_deg": 118,
  "distance_mm": 380
}

```

The backend must preserve both angle and distance.

A fixed tof_front_mm-only abstraction is insufficient for the scanning architecture.

8. MPU6050 CONTRACT

8.1 Purpose

The MPU6050 is primarily used for:

- heading assistance;
- turn completion;
- orientation estimation.

It is not a guarantee of:

- accurate global X/Y;
- full inertial navigation;
- full SLAM;
- encoder-equivalent odometry.

8.2 Initial behavior

Brain:

1. initialize MPU6050;
2. read accelerometer and gyro;
3. calibrate gyro bias while stationary;
4. integrate gyro Z for heading assistance;
5. expose telemetry.

Backend:

- validate health;
- track staleness;
- use heading for orientation/turn assistance.

8.3 Prohibited

Do not double-integrate accelerometer readings and claim accurate position.

8.4 Encoders

Wheel encoders are not present in the current physical baseline.

Do not create encoder-dependent functionality that pretends encoder telemetry exists.

Future scope may add wheel encoders and RTK-GPS.

9. LOCALIZATION AND MAPPING

Coordinate frame:

+X = forward
+Y = left
+Z = up

Yaw:

- counter-clockwise from +X;
- normalized to $[-\pi, \pi]$;
- radians internally.

Existing occupancy grid:

- 200 × 200 cells;
- 0.05 m/cell;
- origin approximately [-5.0 m, -5.0 m];
- free = 0;
- occupied = 100;
- unknown = -1;
- Bresenham-style raycasting.

For a stationary rover:

```
x = distance_m × cos(angle_rad)
y = distance_m × sin(angle_rad)
```

For moving global mapping, valid pose is required.

Do not claim globally accurate persistent SLAM without a validated pose source.

10. NAVIGATION AUTHORITY

10.1 Absolute rule

Only one layer may control movement at any moment.

10.2 Initial prototype

The existing ARACHNID movement/navigation behavior remains the physical low-level authority where required by the tested platform.

AESAR mission logic requests high-level station movement.

It must not directly manipulate motors.

10.3 Forbidden paths

```
Vision → Motors
AESA → Motors
Frontend → Motors
Mission FSM → Motor GPIO
```

Correct pattern:

```
Mission
↓
High-level navigation request
↓
Integration layer
↓
Existing ARACHNID navigation/control
↓
Drive ESP32
↓
Motors
```

11. SAFETY CONTRACT

11.1 Hardware watchdog

Rover ESP32 watchdog:

- timeout = 1000 ms;
- loss of valid UART commands → motor stop.

Do not disable it.

11.2 Obstacle stop

AESAR specification baseline:

- safety distance = 180 mm / 0.18 m.

Any physical change must be tested and documented.

11.3 E-stop

The backend/frontend E-stop:

- sends STOP;
- latches emergency state;
- prevents normal motion until reset.

11.4 Communication loss

Brain must stop the actuator path when its controlling connection is lost.

Backend must also issue STOP on applicable failure/disconnect paths.

11.5 Safety precedence

```
E-STOP / SAFETY
>
MISSION
>
NAVIGATION
>
NORMAL MOTION
```

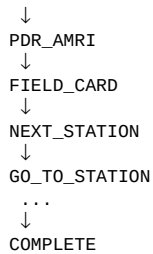
Safety must never wait for:

- YOLO;
- PDR;
- AMRI;
- UI rendering.

12. MISSION STATE MACHINE

Canonical high-level lifecycle:

```
IDLE
↓
PLAN_MISSION
↓
GO_TO_STATION
↓
ARRIVED
↓
CAPTURE
↓
YOLO
↓
ABIOTIC_SAMPLE
```



The source AESAR specification expresses the same lifecycle as:

- standby/diagnostics;
- row following;
- station detection;
- dwell/capture;
- YOLO;
- abiotic sampling;
- increment station;
- final field card after station 10.

12.1 Mission size

Exactly:

10 representative stations

Stations:

1 ... 10

12.2 Canopy views

Target:

- lower;
- middle;
- upper.

If a view cannot be captured, mark it unavailable. Do not fabricate it.

12.3 Station completion

A station is complete only after:

- arrival/dwell;
- required image attempts;
- vision result or explicit failure;
- abiotic sample or explicit unavailable status;
- station record.

13. PERSON 2 — VISION/AI CONTRACT

Branch:

feature/vision-ai

Owns:

- camera abstraction;
- image capture;
- preprocessing;
- YOLO;
- pest detection;
- defender detection;
- vision result schema;
- tests;
- mock images.

Pest classes specified:

- Brown Planthopper (BPH);
- Green Leafhopper (GLH);
- Yellow Stem Borer;
- Cotton Aphids;
- Chilli Thrips;
- American Bollworm.

Defender classes:

- Ladybird Beetle adult/larva;
- Lycosa/Wolf Spider;
- Mirid Bug;
- Carabid Beetle;
- Green Lacewing.

13.1 Vision result

```
{
  "station_id": 3,
  "view": "upper",
  "detections": [
    {
      "class": "aphid",
      "category": "pest",
      "confidence": 0.91,
      "count": 1
    }
  ]
}
```

Do not claim a model accuracy or training dataset result unless the team actually validates it.

14. PERSON 3 — PDR/AMRI/AESA CONTRACT

Branch:

feature/ecosystem-intelligence

14.1 PDR

$$\text{PDR} = \frac{\sum(\text{wp}, j \times \text{Npest}, i, j)}{(\sum(\text{wd}, k \times \text{Ndefender}, i, k) + \epsilon)}$$

where:

- $N_{pest,i,j}$ = pest count;
- w_p,j = pest voracity weight;
- $N_{defender,i,k}$ = defender count;
- w_d,k = defender predation weight;
- $\varepsilon = 10^{-3}$.

If species-specific weights are not explicitly provided, they are TBD.

Do not invent them.

14.2 AMRI

$$\begin{aligned} \text{AMRI}_i = & 0.35 \times ((T_{\text{canopy},i} - T_{\text{opt}}) / T_{\text{opt}})^2 \\ & + 0.40 \times (RH_{\text{ambient},i} / 100) \\ & + 0.25 \times (1 - M_{\text{soil},i} / M_{\text{field_cap}}) \end{aligned}$$

Variables:

- T_{canopy} = canopy temperature;
- T_{opt} = optimum temperature;
- RH_{ambient} = relative humidity percentage;
- M_{soil} = soil moisture;
- $M_{\text{field_cap}}$ = field soil capacity.

If crop-specific T_{opt} or $M_{\text{field_cap}}$ is not provided, configuration is required before claiming a calibrated AMRI.

15. NPM DECISION MATRIX

Condition	Ecosystem state	Chemical recommendation	NPM intervention
$PDR \leq 1.0$ AND $AMRI < 0.50$	Perfect Ecological Balance	STRICTLY FORBIDDEN (0% spray)	Natural biological control; no intervention required
$1.0 < PDR \leq 2.5$ AND $AMRI < 0.65$	Moderate Pest Buildup	NO SYNTHETIC CHEMICALS	Yellow/blue sticky traps; Trichogramma egg parasitoids; NSKE 5%
$PDR > 2.5$ OR $AMRI \geq 0.80$	Ecological Imbalance	Non-Pesticidal Corrective Focus	Bird perches; botanical bio-extracts; irrigation adjustment; re-survey in 72 h

15.1 Undefined decision region

The supplied matrix does not explicitly define every combination between:

- $AMRI$ 0.65 and 0.80;
- $PDR > 1.0$ with $AMRI \geq 0.65$ and < 0.80 .

Therefore the engine must return:

UNSPECIFIED / REVIEW_REQUIRED

unless the team formally adds a documented rule.

16. FIELD CARD CONTRACT

A station field card may contain:

- station ID;

- coordinates;
- view status;
- pest detections;
- defender detections;
- counts;
- PDR;
- soil moisture;
- canopy temperature;
- relative humidity;
- AMRI;
- ecosystem state;
- recommendation;
- timestamp;
- data-quality/confidence information.

No card may claim a measurement that was not actually obtained.

17. PERSON 4 — MISSION/DASHBOARD CONTRACT

Branch:

`feature/mission-dashboard`

Owns:

- mission FSM;
- station manager;
- 10-station lifecycle;
- mission status;
- dashboard;
- live telemetry visualization;
- station UI;
- field card UI.

Mission module must communicate through high-level contracts.

It must not:

- access GPIO;
- access ESP32 directly;
- issue raw motor commands.

18. PERSON 1 — HARDWARE/INTEGRATION CONTRACT

Branch:

`feature/hardware-integration`

Owns:

- Brain ESP32;

- MPU6050;
- ToF/servo;
- ultrasonic integration;
- agricultural sensor hardware;
- camera hardware connection;
- hardware gateway;
- telemetry normalization;
- localization/heading;
- mapping integration;
- safety;
- ARACHNID integration;
- final end-to-end physical testing.

Person 1 is final integration owner.

19. FILE OWNERSHIP

firmware/brain/	Person 1
backend/app/hardware/	Person 1
backend/app/localization/	Person 1
backend/app/safety/	Person 1
backend/app/mapping/	Person 1 + Person 4
backend/app/vision/	Person 2
backend/app/ecosystem/	Person 3
backend/app/decision/	Person 3
backend/app/mission/	Person 4
frontend/	Person 4
backend/app/models/	Shared / integration reviewed
docs/	Shared / integration reviewed
tests/	Subsystem owner
data/mock/	Shared

20. GIT CONTRACT

Branch structure:

```

main
|
■■■ integration
    |
    ■■■ feature/hardware-integration
    ■■■ feature/vision-ai
    ■■■ feature/ecosystem-intelligence
    ■■■ feature/mission-dashboard
  
```

Rules:

- no direct push to main;
- feature branches originate from integration;
- PRs target integration;
- integration is tested;
- only stable/demo-ready code reaches main;
- never branch from another person's feature branch.

Commands:

```

git clone <REPOSITORY_URL>
cd AESAR
  
```

```
git fetch origin
git checkout integration
git pull origin integration
```

Person 2:

```
git checkout -b feature/vision-ai
git push -u origin feature/vision-ai
```

Person 3:

```
git checkout -b feature/ecosystem-intelligence
git push -u origin feature/ecosystem-intelligence
```

Person 4:

```
git checkout -b feature/mission-dashboard
git push -u origin feature/mission-dashboard
```

Feature workflow:

```
feature
↓
commit
↓
push
↓
Pull Request
↓
integration
↓
test
↓
main
```

Recommended commit prefixes:

```
feat:
fix:
docs:
refactor:
test:
chore:
```

21. INTERFACE CHANGE CONTROL

Any change to:

- JSON fields;
- units;
- enum/state names;
- REST endpoints;
- WebSocket events;
- command format;
- required/optional fields;
- safety thresholds;
- coordinate convention

must update the relevant documentation and notify Person 1.

No silent interface changes.

22. MOCK-FIRST DEVELOPMENT

Required:

```
data/mock/  
■■■ telemetry.json  
■■■ detections.json  
■■■ ecosystem.json  
■■■ mission.json
```

Mock data must use the same logical contracts as real data.

Teammates must be able to develop without physical rover access.

23. FRONTEND CONTRACT

Frontend communicates only with FastAPI.

Frontend must not:

- parse raw ESP32 packets;
- translate motor commands;
- perform hardware coordinate transforms;
- implement hardware safety;
- connect directly to ESP32.

Existing map components should be reused where possible.

24. EXPECTED EVENTS

The normalized backend may expose:

```
sensor_telemetry  
rover_status  
map_update  
navigation_update  
exploration_update  
tof_scan  
aesa_transect_packet  
field_card
```

Exact final schemas must be documented before integration.

25. TEST CONTRACT

Minimum physical tests:

1. Brain/backend connection.
2. ToF sweep.
3. Five ultrasonic telemetry.
4. MPU6050 heading.
5. Manual FORWARD/BACKWARD/LEFT/RIGHT/STOP.
6. Rover watchdog.
7. Obstacle stop.
8. Wi-Fi/TCP loss stop.
9. Stationary mapping.
10. End-to-end mission.

Software tests:

Vision

- empty detections;
- multiple detections;
- confidence;
- schema validation;
- unavailable image.

PDR

- zero pests;
- zero defenders;
- multiple species;
- epsilon;
- missing data.

AMRI

- normal values;
- boundaries;
- missing values;
- crop configuration.

Decision engine

- every defined boundary;
- undefined decision region;
- no silent recommendation.

Mission

- mission start;
- station 1;
- station 10;
- capture failure;
- vision failure;
- sensor failure;
- safety stop;
- completion;
- reset.

Frontend

- connection;
- telemetry;
- mission progress;
- station result;
- field card;
- E-stop;
- stale data.

26. END-TO-END ACCEPTANCE

```
START
↓
Diagnostics
↓
Plan 10 stations
↓
Navigate
↓
Arrive
↓
Capture lower/middle/upper
↓
YOLO
↓
Abiotic sample
↓
PDR + AMRI
↓
AESA decision
↓
Field card
↓
Next station
↓
...
↓
Station 10
↓
Final field card
↓
COMPLETE
```

The demo must visibly establish:

- rover movement;
- obstacle safety;
- sensor telemetry;
- station progress;
- vision output;
- PDR;
- AMRI;
- recommendation;
- field card;
- mapping/position information only to the level physically justified.

27. DEMO CHECKLIST

Before autonomous movement:

1. power/battery;
2. motor wiring;
3. wheel/mechanical inspection;
4. Brain power;
5. Rover power;
6. UART;
7. Wi-Fi;
8. telemetry;
9. E-stop;

10. watchdog;
11. obstacle stop;
12. camera;
13. mission reset;
14. one-station test;
15. multi-station test;
16. full mission.

Do not debug new firmware while the rover is moving.

28. OUT OF SCOPE FOR THE INITIAL HACKATHON

Do not make these critical-path dependencies:

- wheel encoder installation;
- RTK-GPS;
- full visual SLAM;
- full inertial dead reckoning;
- ROS2 migration;
- Nav2 migration;
- Drive ESP32 replacement;
- new velocity motor protocol;
- chassis redesign;
- chemical spraying hardware;
- multispectral sensing.

These are future extensions unless formally added by contract revision.

29. KNOWN LIMITATIONS

Because the current physical platform lacks wheel encoders:

- global position can drift;
- MPU6050 improves heading but does not guarantee accurate X/Y;
- mapping quality depends on pose quality;
- prototype station coordinates may be approximate.

The team must not claim more than has been physically validated.

30. DEFINITION OF DONE

A subsystem is complete only when:

- interface is documented;
- tests exist;
- mock data works;

- errors are handled;
- missing data is explicit;
- unrelated subsystems remain functional;
- feature branch is pushed;
- PR is created;
- integration impact is documented;
- integration checks pass.

31. FINAL NON-NEGOTIABLE RULES

1. Do not assume.
2. Do not silently change interfaces.
3. Do not bypass safety.
4. Only one movement authority at a time.
5. Current motor protocol is ASCII, not JSON.
6. Wheel encoders are not present.
7. MPU6050 is heading assistance, not full SLAM.
8. Browser never connects directly to ESP32.
9. Feature branches never push directly to main.
10. Unknown values become TBD, not invented values.

32. QUICK REFERENCE

REPOSITORY
AESAR

TEAM
P1 = Hardware + Integration
P2 = Vision / AI
P3 = PDR + AMRI + AESA
P4 = Mission + Dashboard

BRANCHES
main
integration
feature/hardware-integration
feature/vision-ai
feature/ecosystem-intelligence
feature/mission-dashboard

CONTROLLERS
Brain ESP32 #2
Rover ESP32 #1

BRAIN → BACKEND
TCP :80
JSON telemetry

BRAIN → ROVER
UART 115200
ASCII commands

COMMANDS
FORWARD
BACKWARD
LEFT
RIGHT
STOP

WATCHDOG
1000 ms

OBSTACLE BASELINE

180 mm

TOF
VL53L0X
Servo
30°-150°
~2° steps

I2C
SDA 21
SCL 22

MPU6050
Heading assistance

ENCODERS
NOT PRESENT

MAP
200×200
0.05 m/cell
0 free
100 occupied
-1 unknown

MISSION
10 stations

VIEWS
LOWER / MIDDLE / UPPER

PDR EPSILON
 10^{-3}

AMRI
 $\alpha=0.35$
 $\beta=0.40$
 $\gamma=0.25$

FINAL INTEGRATION OWNER
Person 1

END OF AESAR MASTER IMPLEMENTATION CONTRACT v2.1